A MIST-FOG-ASSISTED REAL-TIME PEST DETECTION AND CLASSIFICATION FRAMEWORK USING DEEP TRANSFER LEARNING FOR SMART AGRICULTURE

A MAJOR PROJECT REPORT
SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENT
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Declaration

This is to certify that:

- The work contained in this project report is original and has been done by me under the supervision of my designated supervisor, Dr. Dilip Senapati.
- The work has not been submitted to any other Institute or University for any degree, diploma, or academic distinction.
- I have conformed to the norms, regulations, and guidelines prescribed in the Ethical Code of Conduct of the Department and University.
- Whenever I have used material (data, visual images, theoretical formulations, and code sequences) from external sources, I have given due credit and acknowledgment to them by appropriate citations in the text of the report and listed them in the references.

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This is to certify that the project report titled "A Mist-Fog-Assisted Real-Time Pest Detection and Classification Framework Using Deep Transfer Learning for Smart Agriculture", submitted to the Department of Computer Science, University of Delhi, by Sujeet Kumar (Roll No. 24234747005), has been carried out under my direct supervision. This work is done in partial fulfillment of the requirements for the completion of the Master of Science (M.Sc.) in Computer Science.

It is further certified that this work is original, has been carried out under my guidance, and to the best of my knowledge, has not been submitted previously for the award of any other degree or diploma.

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Abstract

The global agricultural sector faces unprecedented challenges driven by population growth and ecological volatility. Automated pest monitoring has emerged as a crucial approach to preserve crop yields and reduce broad-spectrum chemical pesticide use. However, traditional cloud architectures introduce severe communication delays, high bandwidth costs, and network dependencies that limit real-time field deployment. This project report presents a holistic distributed computing framework that operates natively at the farm boundary by leveraging a hierarchical Mist-Fog-Cloud topology. The proposed system deploys an explicitly optimized MobileNetV3-Large core directly onto localized edge workstations to achieve low-latency, autonomous inference. To handle the complex intra-class similarities and severe dataset skews inherent to agricultural tracking, a dynamic weighted random sampling routine is integrated into the data execution loop, balancing gradient descent updates across 9 distinct insect categories (aphids, armyworm, beetle, bollworm, grasshopper, mites, mosquito, sawfly, and stem borer).

Experimental evaluations on resource-constrained embedded hardware targets demonstrate that the framework achieves a high top-1 validation accuracy of 97.33% while requiring only 0.62 GFLOPs of computational complexity and a compact 16.4 megabyte storage footprint. By processing visual data packages within 23.5 milliseconds, the system delivers immediate localized diagnostics independent of external internet connectivity, establishing a highly generalizable strategy for smart farming infrastructure. Furthermore, a comparative benchmarking analysis demonstrates that our optimized MobileNetV3-Large core achieves the best operational balance, outperforming larger architectures like ResNet18, EfficientNet-B0, and EfficientNetV2-S in resource utilization, memory consumption, and inference latency, making it the most suitable model for direct physical integration into automated pheromone traps and mobile agricultural inspection robots.

Keywords: Smart Agriculture 4.0, Deep Transfer Learning, Mist Computing, Edge Intelligence, Pest Detection, MobileNetV3-Large, Weighted Random Sampler.

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List of Abbreviations

AI	Artificial Intelligence
AMP	Automated Mixed Precision
API	Application Programming Interface
CAGR	Compound Annual Growth Rate
CNN	Convolutional Neural Network
CPU	Central Processing Unit
DL	Deep Learning
DUCS	Department of Computer Science, University of Delhi
FIFO	First-In-First-Out
FLOPs	Floating Point Operations
FPS	Frames Per Second
GFLOPs	Giga Floating Point Operations
GPU	Graphics Processing Unit
IDS	Intrusion Detection System
IoT	Internet of Things
LR	Learning Rate
MACs	Multiply-Accumulate Operations
mAP	mean Average Precision
ML	Machine Learning
QoI	Quality of Information
RAM	Random Access Memory
ReLU	Rectified Linear Unit
RGB	Red-Green-Blue
SBC	Single-Board Computer
TOC	Table of Contents
UC	University Course / Core
WAN	Wide Area Network